

5.5 For each of the following two time-series regression models, and assuming MR1-MR6 hold, find $\text{var}(b_2|\mathbf{x})$ and examine whether the least squares estimator is consistent by checking whether $\lim_{T \rightarrow \infty} \text{var}(b_2|\mathbf{x}) = 0$.

- a. $y_t = \beta_1 + \beta_2 t + e_t, t = 1, 2, \dots, T$. Note that $\mathbf{x} = (1, 2, \dots, T)$, $\sum_{t=1}^T (t - \bar{t})^2 = \sum_{t=1}^T t^2 - \left(\sum_{t=1}^T t\right)^2 / T$, $\sum_{t=1}^T t = T(T+1)/2$ and $\sum_{t=1}^T t^2 = T(T+1)(2T+1)/6$.
- b. $y_t = \beta_1 + \beta_2 (0.5)^t + e_t, t = 1, 2, \dots, T$. Here, $\mathbf{x} = (0.5, 0.5^2, \dots, 0.5^T)$. Note that the sum of a geometric progression with first term r and common ratio r is

$$S = r + r^2 + r^3 + \dots + r^n = \frac{r(1 - r^n)}{1 - r}$$

- c. Provide an intuitive explanation for these results.

$$a. \text{var}(b_2|\mathbf{x}) = \frac{\sigma^2}{\sum_{t=1}^T (x_t - \bar{x})^2}$$

$$\sum (t - \bar{t})^2 = \sum t^2 - \frac{(\sum t)^2}{T} \quad \text{代 } \sum t = \frac{T(T+1)}{2} \quad \text{和 } \sum t^2 = \frac{T(T+1)(2T+1)}{6}$$

$$\Rightarrow \sum (t - \bar{t})^2 = \frac{T(T+1)(2T+1)}{6} - \left[\frac{T(T+1)}{2}\right]^2 \div T = \frac{T^3 - T}{12}$$

$$\text{代回 } \text{var}(b_2|\mathbf{x}) \Rightarrow \frac{\sigma^2}{\sum (t - \bar{t})^2} = \frac{12\sigma^2}{T^3 - T}$$

$$\lim_{T \rightarrow \infty} \text{var}(b_2|\mathbf{x}) = 0 \Rightarrow \text{OLS 一致} \quad \text{✗}$$

$$b. x_t = (0.5)^t, \rightarrow 0.$$

$$\sum x_t = \frac{0.5(1 - 0.5^T)}{1 - 0.5} = 1 - 0.5^T, \quad \bar{x} = \frac{1}{T} \sum x_t = \frac{1}{T} \sum (0.5)^t = \frac{1 - 0.5^T}{T}$$

$$\sum x_t^2 = \frac{0.25(1 - 0.25^T)}{1 - 0.25} = \frac{1}{2}(1 - 0.25^T)$$

$$\sum (x_t - \bar{x})^2 = \sum x_t^2 - \frac{(\sum x_t)^2}{T} = \frac{1}{2}(1 - 0.25^T) - \frac{(1 - 0.5^T)^2}{T}$$

$$\text{var}(b_2|\mathbf{x}) = \frac{\sigma^2}{\frac{1}{2}(1 - 0.25^T) - \frac{(1 - 0.5^T)^2}{T}}$$

$$\lim_{T \rightarrow \infty} \text{var}(b_2|\mathbf{x}) = \lim_{T \rightarrow \infty} \frac{\sigma^2}{\frac{1}{2}(1 - 0.25^T) - \frac{(1 - 0.5^T)^2}{T}} = \frac{\sigma^2}{\frac{1}{2} \times 1 - 0} = 2\sigma^2 \neq 0 \Rightarrow \text{OLS 不一致} \quad \text{✗}$$

c. (a) 小題中, T 增加會提供更多 β_2 以及 x 和 y 關係的資訊。

(b) 小題中, $x_t = (0.5)^t$ 在 T 大小適中時就為 0, 未提供更多 β_2 有關的資訊。✗

5.6 Suppose that, from a sample of 63 observations, the least squares estimates and the corresponding estimated covariance matrix are given by

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} \quad \widehat{\text{cov}}(b_1, b_2, b_3) = \begin{matrix} & \begin{matrix} b_1 & b_2 & b_3 \end{matrix} \\ \begin{matrix} b_1 \\ b_2 \\ b_3 \end{matrix} & \begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix} \end{matrix}$$

Using a 5% significance level, and an alternative hypothesis that the equality does not hold, test each of the following null hypotheses:

- $\beta_2 = 0$
- $\beta_1 + 2\beta_2 = 5$
- $\beta_1 - \beta_2 + \beta_3 = 4$

都使用 two-tail test. critical value: $t_{(5\%, 63-3)} = -2$, $t_{(1-5\%, 63-3)} = 2$

$\Rightarrow$ rejection region: $t < -2$ 或 $t > 2$

a. $H_0: \beta_2 = 0$, $H_1: \beta_2 \neq 0$.

$$\text{test statistic: } t = \frac{b_2}{\text{se}(b_2)} = \frac{3}{\sqrt{4}} = 1.5$$

$\because -2 < t = 1.5 < 2 \Rightarrow$ 无法拒绝 $H_0 \Rightarrow$ there's no sample evidence to suggest that $\beta_2 \neq 0$

b. $H_0: \beta_1 + 2\beta_2 = 5$, $H_1: \beta_1 + 2\beta_2 \neq 5$

$$\text{se}(b_1 + 2b_2) = \sqrt{\widehat{\text{var}}(b_1 + 2b_2)} = \sqrt{\widehat{\text{var}}(b_1) + 2^2 \widehat{\text{var}}(b_2) + 2 \times 2 \text{cov}(b_1, b_2)} = \sqrt{3 + 4 \times 4 + 4 \times (-2)} = \sqrt{11} = 3.3166$$

$$\text{test statistic: } t = \frac{(b_1 + 2b_2) - 5}{\text{se}(b_1 + 2b_2)} = \frac{(2 + 2 \times 3) - 5}{3.3166} = 0.9045$$

$\because -2 < t = 0.9045 < 2 \Rightarrow$ 无法拒绝 $H_0 \Rightarrow$ there's no sample evidence to suggest that $\beta_1 + 2\beta_2 \neq 5$.

c. $H_0: \beta_1 - \beta_2 + \beta_3 = 4$, $H_1: \beta_1 - \beta_2 + \beta_3 \neq 4$

$$\text{se}(b_1 - b_2 + b_3) = \sqrt{\widehat{\text{var}}(b_1 - b_2 + b_3)} = \sqrt{\widehat{\text{var}}(b_1) - \widehat{\text{var}}(b_2) + \widehat{\text{var}}(b_3) - 2 \text{cov}(b_1, b_2) + 2 \text{cov}(b_1, b_3) - 2 \text{cov}(b_2, b_3)} = \sqrt{3 + 4 + 3 - 2 \times (-2) + 2 \times 1 - 2 \times 0} = \sqrt{16} = 4$$

$$\text{test statistic: } t = \frac{(b_1 - b_2 + b_3) - 4}{\text{se}(b_1 - b_2 + b_3)} = \frac{2 - 3 + (-1) - 4}{4} = -1.5$$

$\because -2 < t = -1.5 < 2 \Rightarrow$ 无法拒绝 $H_0 \Rightarrow$ there's no sample evidence to suggest that $\beta_1 - \beta_2 + \beta_3 \neq 4$

5.20 A generalized version of the estimator for β_2 proposed in Exercise 2.9 by Professor I.M. Mean for the regression model $y_i = \beta_1 + \beta_2 x_i + e_i$, $i = 1, 2, \dots, N$ is

$$\hat{\beta}_{2,mean} = \frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1}$$

where $(\bar{y}_1, \bar{x}_1)$ and $(\bar{y}_2, \bar{x}_2)$ are the sample means for the first and second halves of the sample observations, respectively, after ordering the observations according to increasing values of x . Given that assumptions MR1–MR6 hold:

- Show that $\hat{\beta}_{2,mean}$ is unbiased.
- Derive an expression for $\text{var}(\hat{\beta}_{2,mean} | \mathbf{x})$.
- Write down an expression for $\text{var}(\hat{\beta}_{2,mean})$.
- Under what conditions will $\hat{\beta}_{2,mean}$ be a consistent estimator for β_2 ?
- Randomly generate observations on x from a uniform distribution on the interval (0,10) for sample sizes $N = 100, 500, 1000$, and, if your software permits, $N = 5000$. Assuming $\sigma^2 = 1000$, for each sample size, compute:
 - $\text{var}(b_2 | \mathbf{x})$ and $\text{var}(\hat{\beta}_{2,mean} | \mathbf{x})$ where b_2 is the OLS estimator.
 - Estimates for $E[(s_x^2)^{-1}]$ and $E[4/(\bar{x}_2 - \bar{x}_1)^2]$ where s_x^2 is the sample standard deviation for x using N as a divisor.
- Comment on the relative magnitudes of your answers in part (e), (i) and (ii) and how they change as sample size increases. Does it appear that $\hat{\beta}_{2,mean}$ is consistent?
- Show that $E[(s_x^2)^{-1}] \xrightarrow{p} 0.12$ and $E[4/(\bar{x}_2 - \bar{x}_1)^2] \xrightarrow{p} 0.16$. [Hint: The variance of a uniform random variable defined on the interval (a, b) is $(b - a)^2/12$.]
- Suppose that the observations on x were not ordered according to increasing magnitude but were randomly assigned to any position. Would the estimator $\hat{\beta}_{2,mean}$ be consistent? Why or why not?

a. $\bar{y}_1 = \beta_1 + \beta_2 \bar{x}_1 + \bar{e}_1$

$\bar{y}_2 = \beta_1 + \beta_2 \bar{x}_2 + \bar{e}_2$

$\hat{\beta}_{2,mean} = \frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1} = \frac{\beta_1 + \beta_2 \bar{x}_2 + \bar{e}_2 - \beta_1 - \beta_2 \bar{x}_1 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1} = \beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1}$

$\therefore E(e_i | x) = 0 \Rightarrow E(\bar{e}_1) = E(\bar{e}_2) = 0$

$\Rightarrow E(\hat{\beta}_{2,mean}) = \beta_2 \Rightarrow \hat{\beta}_{2,mean}$ is unbiased

b. $\hat{\beta}_{2,mean} = \beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1}$

$\therefore$ 兩組樣本的大小都是 $\frac{N}{2} \Rightarrow \text{var}(\bar{e}_i) = \frac{\sigma^2}{N/2} = \frac{2\sigma^2}{N}$

$\text{var}(\bar{e}_2 - \bar{e}_1) = \frac{4\sigma^2}{N}$

$\Rightarrow \text{var}(\hat{\beta}_{2,mean} | x) = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}$

c. $\text{var}(\hat{\beta}_{2,mean}) = E\left[\frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2}\right]$
 $= \frac{4\sigma^2}{N} \cdot E\left[\frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\right]$

d. $N \rightarrow \infty$, $\bar{x}_2 - \bar{x}_1$ 不趨近於 0.

若 $(\bar{x}_2 - \bar{x}_1)^2 \xrightarrow{p} C > 0$ (依大數定律)

$\Rightarrow \text{var}(\hat{\beta}_{2,mean}) \rightarrow 0$.

e.

```
> set.seed(123)
> simulate <- function(N, beta1=1, beta2=2, sigma2=1000){
+   x <- runif(N, 0, 10)
+   e <- rmnorm(N, 0, sqrt(sigma2))
+   y <- beta1 + beta2 * x + e
+
+   # 排序
+   order_index <- order(x)
+   x <- x[order_index]
+   y <- y[order_index]
+
+   # 分組
+   half <- N/2
+   x1 <- x[1:half]
+   x2 <- x[half+1:N]
+   y1 <- y[1:half]
+   y2 <- y[half+1:N]
+
+   # mean estimator
+   beta_mean <- (mean(y2) - mean(y1)) / (mean(x2) - mean(x1))
+
+   # OLS
+   model <- lm(y ~ x)
+   beta_ols <- coef(model)[2]
+
+   # variance (given x)
+   var_ols <- sigma2 / sum((x - mean(x))^2)
+   var_mean <- 4 * sigma2 / (N * (mean(x2) - mean(x1))^2)
+
+   # s_x^2
+   sx2 <- mean((x - mean(x))^2)
+
+   return(list(
+     var_b2 = var_ols,
+     var_mean = var_mean,
+     inv_sx2 = 1/sx2,
+     term_mean = 4/(mean(x2) - mean(x1))^2
+   ))
+ }
> Ns <- c(100, 500, 1000, 5000)
> results <- lapply(Ns, function(N){
+   replicate(500, simulate(N), simplify = FALSE)
+ })
>
> summarize <- function(res){
+   var_ols <- mean(sapply(res, function(x) x$var_ols))
+   var_mean <- mean(sapply(res, function(x) x$var_mean))
+   inv_sx2 <- mean(sapply(res, function(x) x$inv_sx2))
+   term_mean <- mean(sapply(res, function(x) x$term_mean))
+
+   return(c(var_ols, var_mean, inv_sx2, term_mean))
+ }
>
> output <- t(sapply(results, summarize))
> colnames(output) <- c("var(b2|x)", "var(betahat2,mean|x)", "E[(sx2)^(-1)]", "E[4/(x2-x1)^2]")
>
> print(output)
      li)      (ii)
var(b2|x) var(betahat2,mean|x) E[(sx2)^(-1)] E[4/(x2-x1)^2]
100  1.22944788      1.65976320    0.1229448    0.1659763
500  0.24113830      0.32226190    0.1205691    0.1611310
1000 0.12035760      0.16078316    0.1203576    0.1607832
5000 0.02399823      0.03199277    0.1199911    0.1599639
```

f. 從(i)可看出: OLS 的 variance 比 mean estimator 的 variance 小, 表示 OLS 更有效率.

從(ii)可看出: $E[(S_x^2)^{-1}] < E[4/(X_2 - X_1)^2]$, 印證(i)的結果.

隨著樣本數 N 上升, 兩種估計量的 variance 都下降

$\Rightarrow \hat{\beta}_{OLS, MEAN}$ is consistent. ✖

g. $Uniform(0,1) \Rightarrow Var(X) = \frac{(10-0)^2}{12} = 8.33$

$$E[(S_x^2)^{-1}] \rightarrow \frac{1}{8.33} = 0.12$$

$$E\left[\frac{4}{(\bar{x}_2 - \bar{x}_1)^2}\right] = 0.16. \quad \text{✖}$$

h. 會讓 $\bar{x}_2 - \bar{x}_1$ 趨近於 0,

造成 $\hat{\beta}_{OLS, MEAN} = \frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1}$ 分母趨近於 0, 使 $\hat{\beta}_{OLS, MEAN}$ 發散而不會 consistent. ✖